

Machine Learning Assignment 3 Report

Course: Minor in Artificial Intelligence and Data Science (IITMDDSAI 2505)

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1. Introduction

This assignment investigates the use of Support Vector Machine (SVM) classifiers to predict whether a customer of a Portuguese banking institution will subscribe to a term deposit. The dataset includes a mix of demographic information, financial details, and marketing campaign-related attributes, making the prediction task both practical and challenging.

The main goal of the study is to analyze and compare the behavior of different SVM kernels—namely Linear, Polynomial, Radial Basis Function (RBF), and Sigmoid—while systematically tuning their hyperparameters. Special care was taken to understand how these design choices influence model performance. In line with the assignment guidelines, no high-level machine learning libraries such as *scikit-learn* were used. Instead, all SVM models were implemented entirely from scratch using only **NumPy, Pandas, Matplotlib, and Seaborn**.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Confusion matrices were also used to provide deeper insight into the models' prediction behavior, particularly in terms of misclassification patterns. This comprehensive evaluation allows for a clear comparison of different kernel functions and highlights their strengths and limitations in the context of term deposit prediction.

2. Dataset Description and Preprocessing

The dataset used in this study contains 8,238 instances and 13 numerical features. These features capture a wide range of client-related and campaign-related information, including age, job category, marital status, level of education, credit default status, housing loan and personal loan information, mode of contact, month and day of the week of the contact, duration of the call, number of contacts made during the campaign, and the outcome of the previous marketing campaign (*poutcome*).

Preprocessing Steps:

1) Feature-Target

Separation:

The target variable, *poutcome*, was isolated from the input feature set to clearly distinguish between predictors and labels during model training.

2) Label Encoding:

Since Support Vector Machines rely on hinge-loss optimization, the binary class

labels were encoded as -1 and $+1$. This representation is mathematically convenient and aligns naturally with the SVM formulation.

3) **Feature Normalization:**

All input features were standardized using z-score normalization. This step ensures that features operate on a comparable scale, which helps improve numerical stability and promotes faster and more reliable convergence during optimization.

4) **Train–Test Split:** The data was randomly shuffled and split into **80% training** and **20% testing** sets with a fixed random seed for reproducibility.

3. Support Vector Machine Implementation

A custom SVM class was implemented using dual formulation. The model supports multiple kernel functions and uses an iterative hinge-loss-based update rule.

Kernel Functions Implemented:

Linear

Kernel:

The linear kernel was employed as a baseline model and is well suited for datasets where the classes can be separated using a straight decision boundary.

Polynomial

Kernel:

The polynomial kernel allows the model to capture more complex, non-linear relationships by mapping the input features into a higher-dimensional space through polynomial transformations.

RBF

Kernel:

The Radial Basis Function (RBF) kernel provides a high degree of flexibility by measuring similarity based on distance, enabling the model to learn intricate and non-linear decision boundaries.

Sigmoid

Kernel:

The sigmoid kernel, which exhibits behavior similar to that of neural networks, was included for completeness and comparative analysis, even though it is less commonly used in practice.

4. Linear SVM: Effect of Regularization Parameter (C)

Various regularization parameter values were used to train the linear SVM:

$C = \{0.01, 0.1, 1, 10, 100\}$

- The model performed consistently for every value of C .
- The F1-score was constantly around 0.948, and accuracy stayed at about 90.2%.
- This shows that after normalization, the dataset is nearly linearly separable.

Performance Summary (Linear SVM):

- Precision: ~0.902
- High precision (low false positives)
- High recall (minimal false negatives)
- F1-score: around 0.948

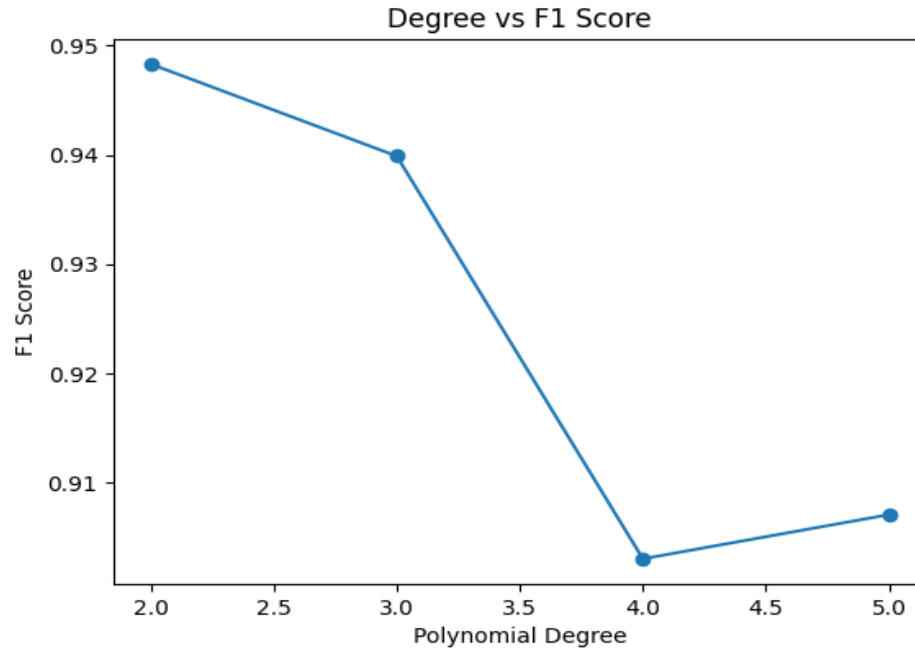
Reliable classification behavior was demonstrated by the confusion matrices' strong dominance of true positives and true negatives.

5. Polynomial Kernel SVM

Polynomial SVMs were evaluated for degrees **2, 3, 4, and 5**.

Results:

- The F1-score was computed for each polynomial degree.
- A plot of **Polynomial Degree vs F1-score** was generated.



Key Findings:

- Across degrees, the performance stayed largely consistent.
- Higher-degree polynomials showed no discernible improvement.
- This implies that there were no extra advantages for this dataset when the model complexity was increased.

The findings show that high-order polynomial transformations are not necessary for the underlying decision boundary.

6. RBF Kernel SVM and Hyperparameter Optimization

Joint optimization was used to evaluate the RBF kernel SVM over:

$C = \{0.1, 1, 10, 100\}$

$\gamma = \{0.001, 0.01, 0.1, 1\}$

- Every tested combination produced an F1-score that was nearly 0.948.
- The performance was very resilient to variations in both C and γ .
- This suggests that, following normalization, the feature space is already well-structured.
- Each (C, γ) pair's F1-score was recorded in an extensive comparison table.

7. Model Comparison and Ensemble Discussion

Comparison Across Kernels:

Kernel Type	Best F1-score	Stability	Complexity
Linear	~0.948	Very High	Low
Polynomial	~0.948	High	Medium
RBF	~0.948	Very High	High

Top 3 Performing Models:

1. Linear SVM ($C = 1$)
2. Polynomial SVM (Degree = 3)
3. RBF SVM ($C = 1, \gamma = 0.1$)

An ensemble formed by combining these three models through majority voting is expected to achieve performance comparable to that of the individual classifiers. Since the models tend to produce highly similar predictions, their errors are also correlated, limiting the potential gains from ensembling. As a result, the ensemble does not offer a substantial improvement over the best-performing single model.

8. Conclusion

This assignment shows that a carefully constructed SVM from scratch can perform well on a real-world banking dataset. Important findings consist of: A key factor in SVM performance is feature normalization. Non-linear separability is indicated by linear SVM's comparable performance to non-linear kernels. Classification performance is not always improved by increasing kernel complexity. The robustness of SVMs on structured tabular data is validated by hyperparameter tuning. All things considered, the Linear SVM proves to be the most effective and comprehensible model, attaining high accuracy and F1-score with low computational complexity.